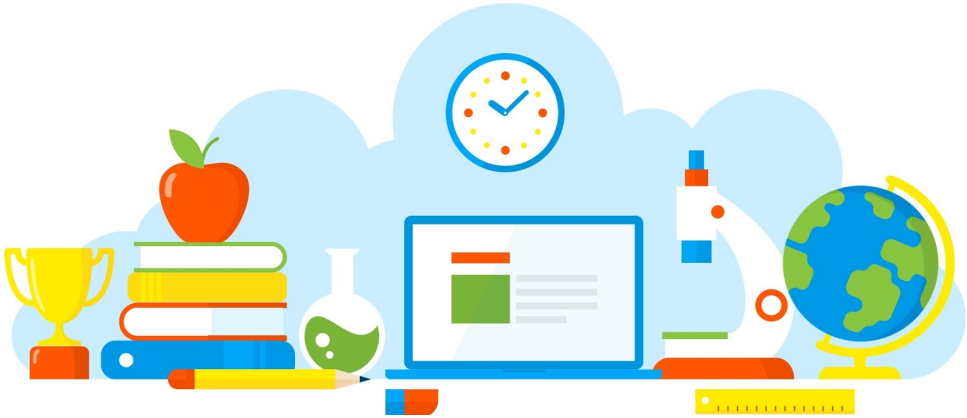
Zoom



COM188 Innovation & Society

Digital Transformation

Prof Maurice Mulvenna

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Outline

- Terminology
- Core gaps
- Transformation
- Not about technology
- Technology focus
- Lo/no code
- State of the art
- National perspective

Terminology

Digital Transformation…

- …the use of new digital technologies such as social media, mobile technology, analytics, or embedded devices to enable major business improvements including enhanced customer experiences, streamlined operations, or new business models.
- …the use of technology to radically improve the performance or reach of enterprises.
- …the realignment of, or new investment in, technology and business models to more effectively engage digital customers at every touch point in the customer experience lifecycle.
- …the combined effects of several digital innovations bringing about novel actors (and actor constellations), structures, practices, values, and beliefs that change, threaten, replace, or complement existing rules of the game within organizations, ecosystems, industries, or fields.

Kraus, S., Jones, P., Kailer, N., Weinmann, A., Chaparro-Banegas, N., & Roig-Tierno, N. (2021). Digital transformation: An overview of the current state of the art of research. Sage Open, 11(3), https://doi.org/10.1177/21582440211047576

Digital Transformation…

- …a consistent networking of all economic sectors and an adaption of actors to new circumstances of the digital economy.
- …an organizational transformation that integrates digital technologies and business processes in a digital economy.
- …comprises the changes associated with the application of digital technology in all aspects of human society.
- …the use of information and communication technology, not when trivial automation is performed, but in the case where fundamentally new capabilities are created in business, public government, and in the lives of people and society.

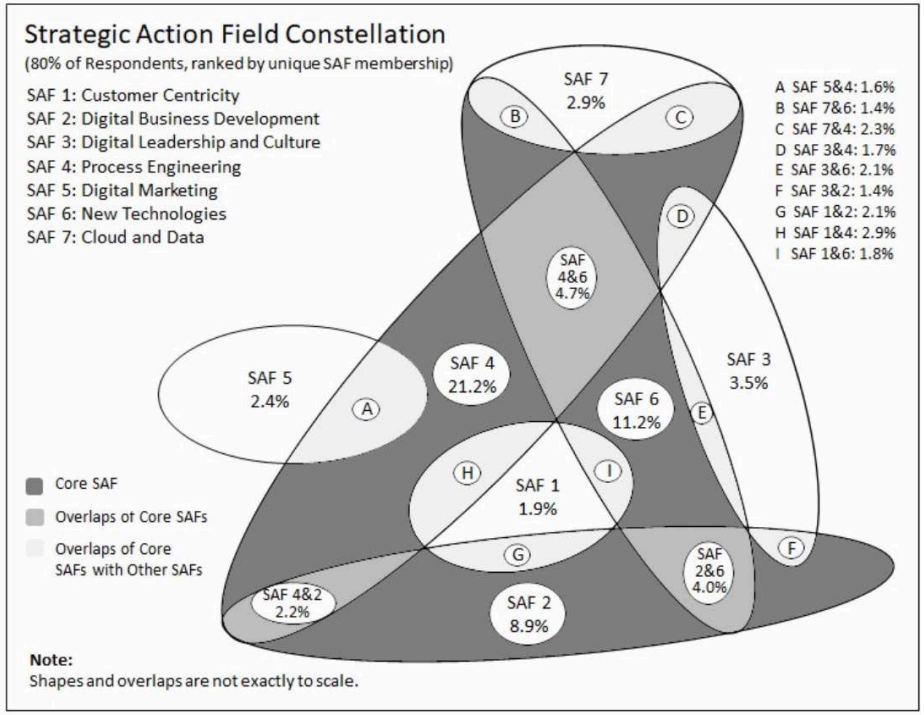
Kraus, S., Jones, P., Kailer, N., Weinmann, A., Chaparro-Banegas, N., & Roig-Tierno, N. (2021). Digital transformation: An overview of the current state of the art of research. Sage Open, 11(3), https://doi.org/10.1177/21582440211047576

Cited keywords associated with 'Digital transformation', from Institute for Scientific Information – Web of Science

Keyword	Record Count	Bar Chart
Digital Transformation	60	
Digitalization	25	
Management	18	
Internet of Things	13	
Internet	10	
Strategy	9	
Government	9	
Industry 4.0	8	
Innovation	8	
Technology	7	
Enterprise Architecture	7	
Competitive Advantage	6	
Information-Technology	6	
Systems	6	
Educational Technology	6	
Digital Business	6	

Reis, J., Amorim, M., Melão, N., & Matos, P. (2018). Digital transformation: a literature review and guidelines for future research. Trends and Advances in Information Systems and Technologies: Volume 1 6, 411-421, https://doi.org/10.1007/978-3-319-77703-0_41

Strategic Action Field (SAF) constellation and membership of Digital Transformation



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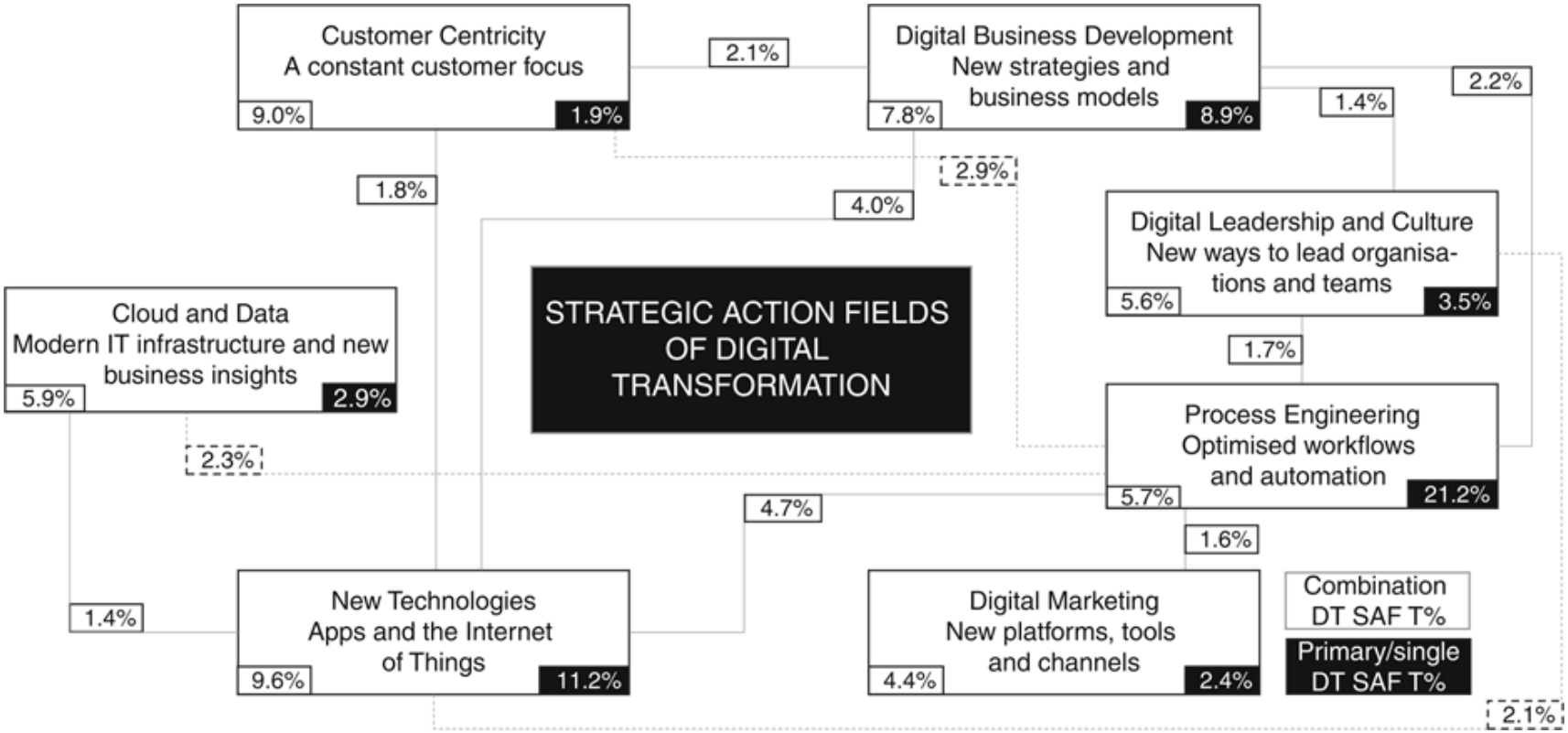


Peter, M. K., Kraft, C., & Lindeque, J. (2020). Strategic action fields of digital transformation: An exploration of the strategic action fields of Swiss SMEs and large enterprises. Journal of Strategy and Management, 13(1), 160-180. https://doi.org/10.1108/JSMA-05-2019-0070

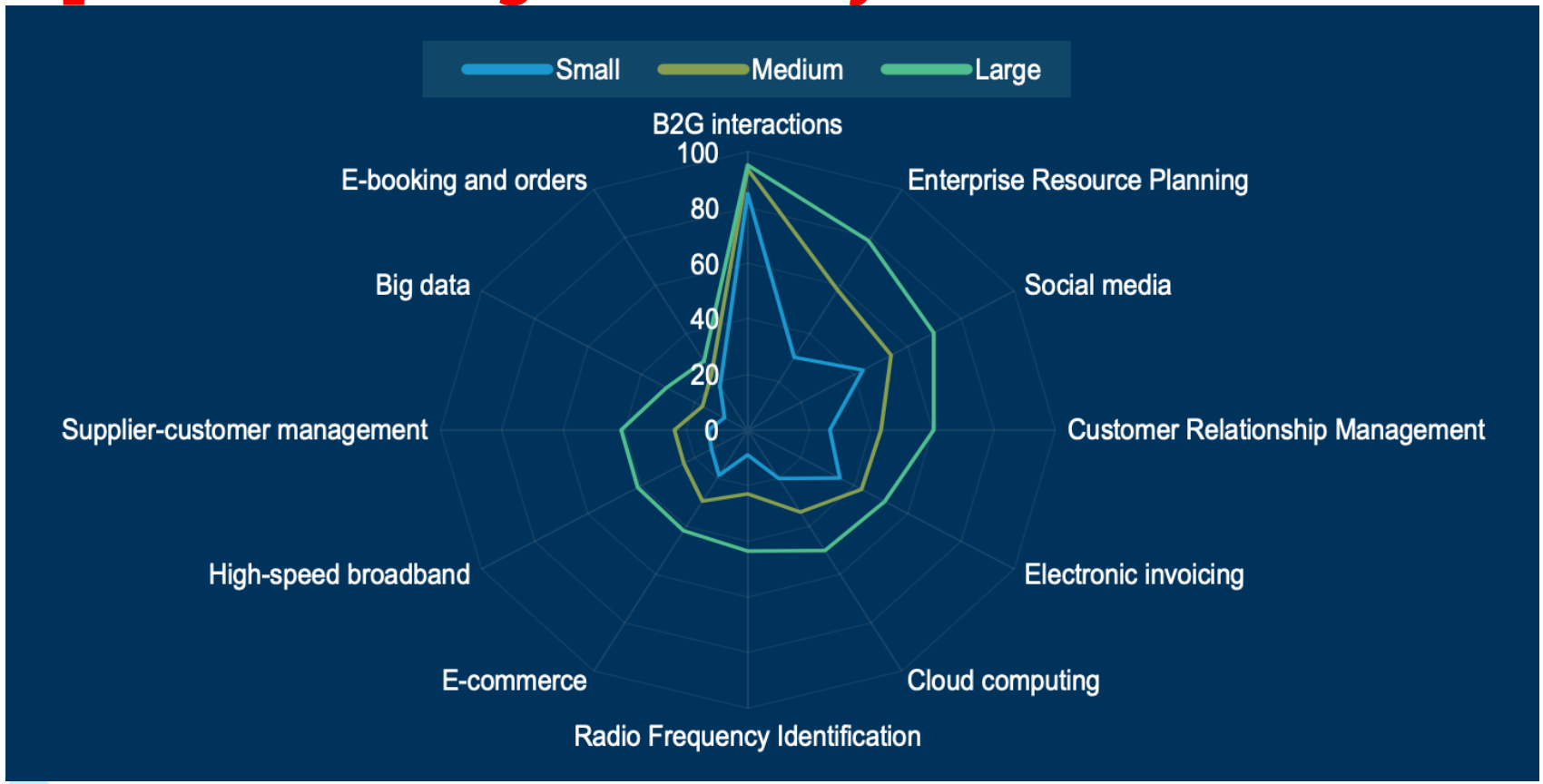
Strategic Action Field (SAF) constellation and membership of Digital Transformation

- Customer Centricity
- Digital Business Development
- Digital Leadership and Culture
- Process Engineering
- Digital Marketing
- New Technologies
- Cloud and Data

Peter, M. K., Kraft, C., & Lindeque, J. (2020). Strategic action fields of digital transformation: An exploration of the strategic action fields of Swiss SMEs and large enterprises. Journal of Strategy and Management, 13(1), 160-180. https://doi.org/10.1108/JSMA-05-2019-0070

Core gaps

ICT Usage by Businesses: SME gaps in adoption are large in many areas



OECD (2015) The OECD Model Survey on ICT Usage by Businesses, https://www.oecd.org/sti/ieconomy/ICT-Model-Survey-Usage-Businesses.pdf

ICT Usage by Businesses: 'Covid-19 crisis has been a game-changer'

- Lockdowns and social distancing imposed a radical rethinking of business models, with firms moving operations online or implementing smart working solutions at short notice in order to remain in business and overcome disruptions in supply chains
- Early evidence from business surveys worldwide point to up to 70% of SMEs having intensified their use of digital technologies due to COVID-19. Many of these changes are poised to last given the investments made and business benefits of the new models.
- Business surveys conducted worldwide over the past months confirm the shift: 75% of the firms surveyed in the United Kingdom have moved to remote working over the period and around a third have invested in new digital capabilities

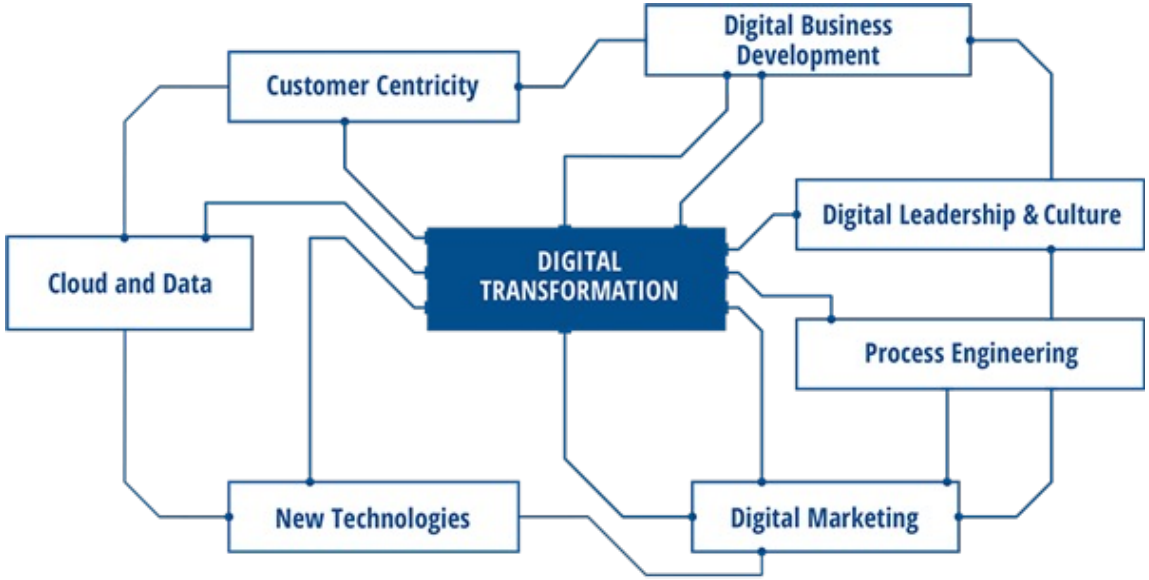
OECD (2021) The Digital Transformation of SMEs: OECD Studies on SMEs and Entrepreneurship, OECD Publishing, Paris, https://doi.org/10.1787/bdb9256a-en

Transformation

What is digital transformation?

https://www.youtube.com/watch?v=GsB1f5uoGWc

Digital Transformation: 7 Action Fields of Transformation



https://marcpeter.com/digital-transformation/

Digital Transformation Canvas
The 7 Action Fields of Transformation

Peter, Marc K (Ed.) (2017): KMU-Transformation: Als KMU die Digitale Transformation erfolgreich umsetzen. Forschungsresultate und Praxisleitfaden. FHNW Hochschule für Wirtschaft, Olten, Switzerland. https://kmu-transformation.ch/ https://digitale-transformation-artikel.ch/wp-content/uploads/2020/07/Digital-Transformation-Canvas-Marc-Peter-English.pdf

How-To Guide for a Digital Transformation

Step 1:

Maturity Analysis

Based on the current situation of the organisation, the maturity analysis is utilised to identify and define the most important drivers and indicators for the transformation project. This enables an initial discussion and the determination of priorities.

Steps 2 and 3:

Strategic Analysis

In a second and third step, the analysis phase deals with the seven action fields and first projects are outlined. From an external perspective (step 2), the customers, the market and new technologies are analysed; from an internal perspective (step 3), processes, the cultural dimensions and leadership approaches, services and data are described. Here, the canvas is used, which provides practical guidance and key questions for workshops.

Step 4:

Strategy Development

The fourth step describes the digital vision and defines the unique value proposition. This includes the development and implementation of digital business options, definition of products and services in the value chain and the required core processes with their IT requirements.

Step 5:

Roadmap & Implementation

The resulting roadmap (the implementation plan) includes the definition of subprojects and project plans in the classical sense. It is recommended to define a feasible project scope, to work with external partners and, where meaningful, to identify quick wins.

Step 6:

Change Management and Leadership

In the sixth step, the key enablers of change management, leadership and cultural development are planned. The development of values and rules of conduct with all employees as well as the coaching and ongoing development of employees are central to a successful change process.

Step 7:

Marketing and Continuous Optimisation

Finally, the renewed organisation and its products and services are marketed. This includes marketing, sales and customer service teams. In addition, market feedback with valuable information flows back into the entire ecosystem in order to continuously identify new potential and improve the value chain. This concludes the process loop of strategic planning.

Marc K Peter (2018) Digital Transformation Canvas: The 7 Action Fields of Transformation, https://digitale-transformation-artikel.ch/wp-content/uploads/2020/07/Digital-Transformation-Canvas-Marc-Peter-English.pdf

Not about technology

Digital Transformation Is Not About Technology

- Companies are pouring millions into “digital transformation” initiatives — but a high percentage of those fail to pay off.
- That’s because companies put the cart before the horse, focusing on a specific technology (“we need a machine-learning strategy!”) rather than doing the hard work of fitting the change into the overall business strategy first.
- Not only should they align tech investments with business goals — they should also lean more on insider knowledge than outside consultants, acknowledge fears about job loss that those insiders may have, develop deep knowledge of how changes will affect customer experience, and use process techniques borrowed from the tech world (experimentation, prototyping, etc.) to facilitate change

Tabrizi, B., Lam, E., Girard, K., & Irvin, V. (2019). Digital transformation is not about technology. Harvard business review, 13(March), 1-6.
https://hbr.org/2019/03/digital-transformation-is-not-about-technology

Digital Transformation Is Not About Technology: Lessons to learn

- **Figure out your business strategy before you invest in anything** – e.g., reduce production lead times, increase speed-to-market, and improve the use of data
- **Leverage insiders** - staff who have intimate knowledge about what works and what doesn't in their daily operations
- **Design customer experience from the outside in** - only way to know where to alter and how to alter is through obtaining extensive and in-depth input from the customers
- **Recognize employees' fear of being replaced** - opportunity for employees to upgrade their expertise to suit the marketplace of the future
- **Bring Silicon Valley start-up culture inside** - agile decision making, rapid prototyping and flat structures

Tabrizi, B., Lam, E., Girard, K., & Irvin, V. (2019). Digital transformation is not about technology. Harvard business review, 13(March), 1-6. https://hbr.org/2019/03/digital-transformation-is-not-about-technology

Digital Transformation Explained

https://www.youtube.com/watch?v=PeD35QUO_Vs

Technology focus

Digital transformation: underlying technologies and platforms

- Digital transformation is a technology-driven continuous change process of companies and our entire society.
- Its cornerstone is ubiquitous embedded computing, connectivity, and flexible value streams.
- Obviously, there's no digital transformation without software.
- But what are the underlying platforms? What lessons can we take from actual projects?

C. Ebert and C. H. C. Duarte, "Digital Transformation," in IEEE Software, vol. 35, no. 4, pp. 16-21, July/August 2018, https://doi.org/10.1109/MS.2018.2801537

Digital transformation: perspectives on technologies

- Early or timely value delivery (agile methods),
- Usage at larger scales (APIs, microservices, and IPv6),
- Applications in new domains (3D modeling and printing, control software, and blockchains), and
- Unpredicted technology combinations (cognitive computing, which combines computer vision, voice recognition, natural-language processing, and machine learning).

C. Ebert and C. H. C. Duarte, "Digital Transformation," in IEEE Software, vol. 35, no. 4, pp. 16-21, July/August 2018, https://doi.org/10.1109/MS.2018.2801537

Digital transformation: technologies

- Collaborative equipment (drones and robots)
- Additive manufacturing and 3D printing
- IoT connected devices
- Agile development
- Blockchain or Hyperledger
- Open APIs and microservices
- AI

C. Ebert and C. H. C. Duarte, "Digital Transformation," in IEEE Software, vol. 35, no. 4, pp. 16-21, July/August 2018, https://doi.org/10.1109/MS.2018.2801537

The Future of Software

Collaboration

- Consumer Internet, social networks, single-customer segmentation, product and service configurators, digital money, computer-assisted collaboration, and crowdsourcing

Convergence

- Mobile enterprises, bioinformatics, the Internet of Things, pervasive sensing, and autonomous systems

Comprehension

- Semantic search, big data handling, smart data, data analytics, data economy, online data validation, and data quality

Cloud

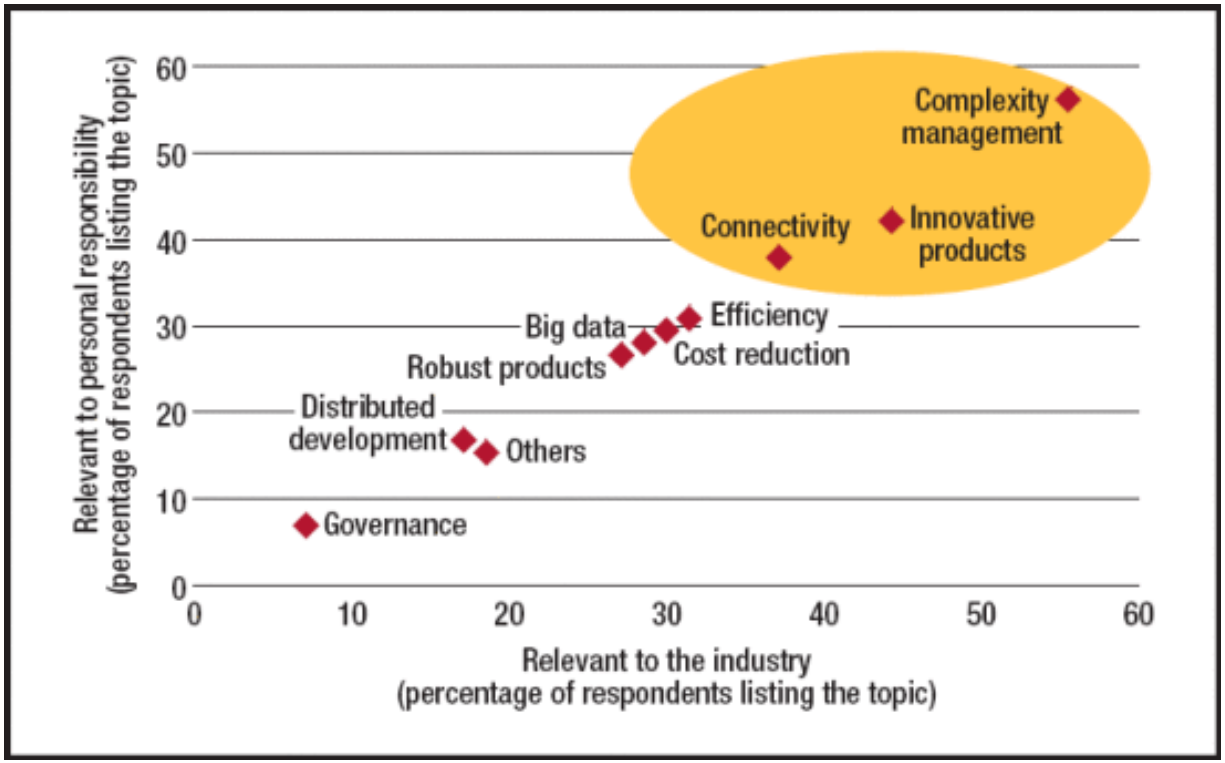
- Applications and services in the cloud, location-based networks, new license models for software and applications, sustainability, and energy efficiency

Connectivity

- Ubiquitous mobile computing, mobile services, cyber-physical systems, Industry 4.0, machine-to-machine communication, sensor networks, and multi-sensor fusion

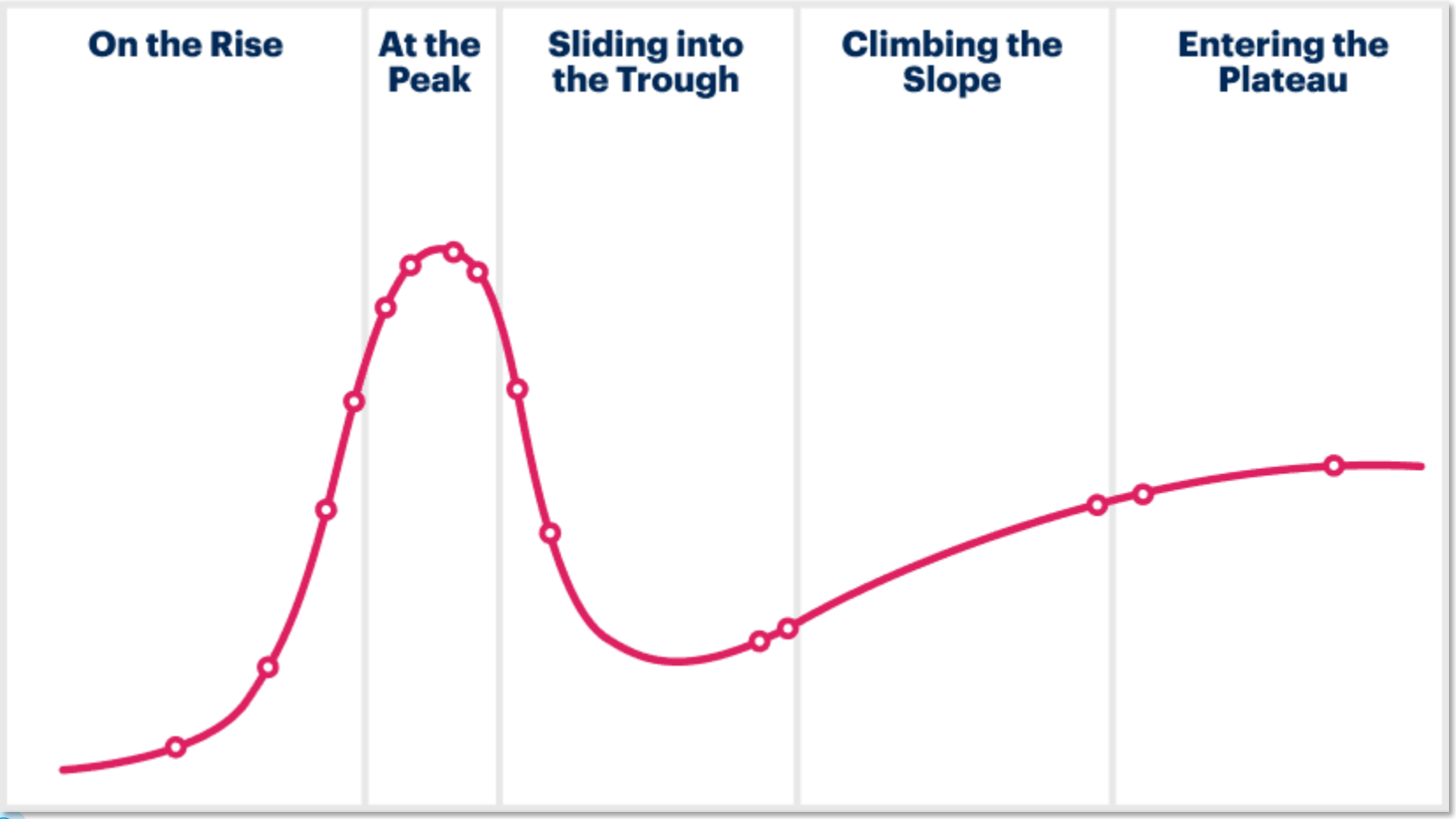
C. Ebert, "Looking into the Future," in IEEE Software, vol. 32, no. 6, pp. 92-97, Nov.-Dec. 2015, https://doi.org/10.1109/MS.2015.142

The Future of Software: industry trends. Innovation, complexity, and connectivity are the major issues for business and technology leaders



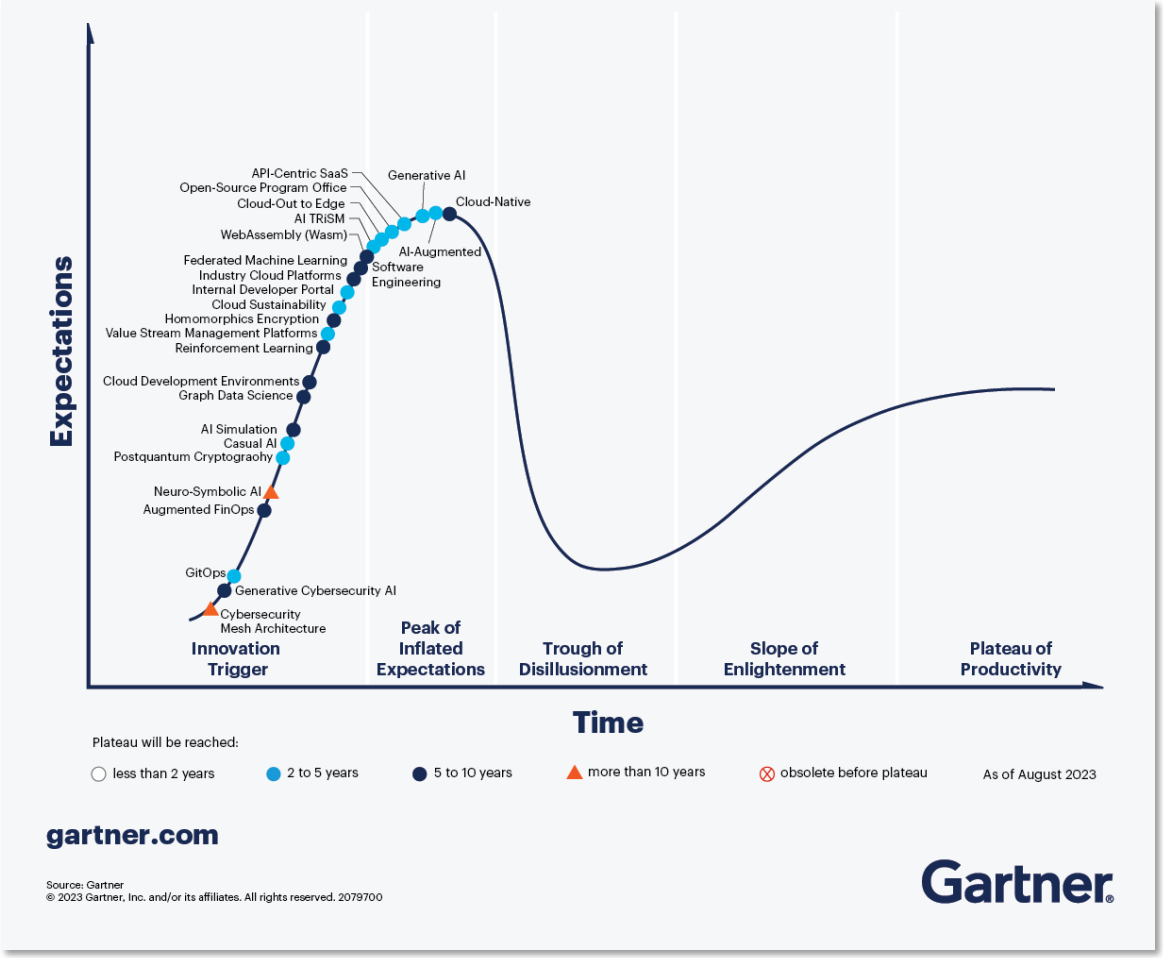
C. Ebert, "Looking into the Future," in IEEE Software, vol. 32, no. 6, pp. 92-97, Nov.-Dec. 2015, https://doi.org/10.1109/MS.2015.142

Gartner Hype Cycle



https://www.gartner.com/en/articles/what-s-new-in-the-2023-gartner-hype-cycle-for-emerging-technologies

Gartner Hype Cycle for Emerging Technologies



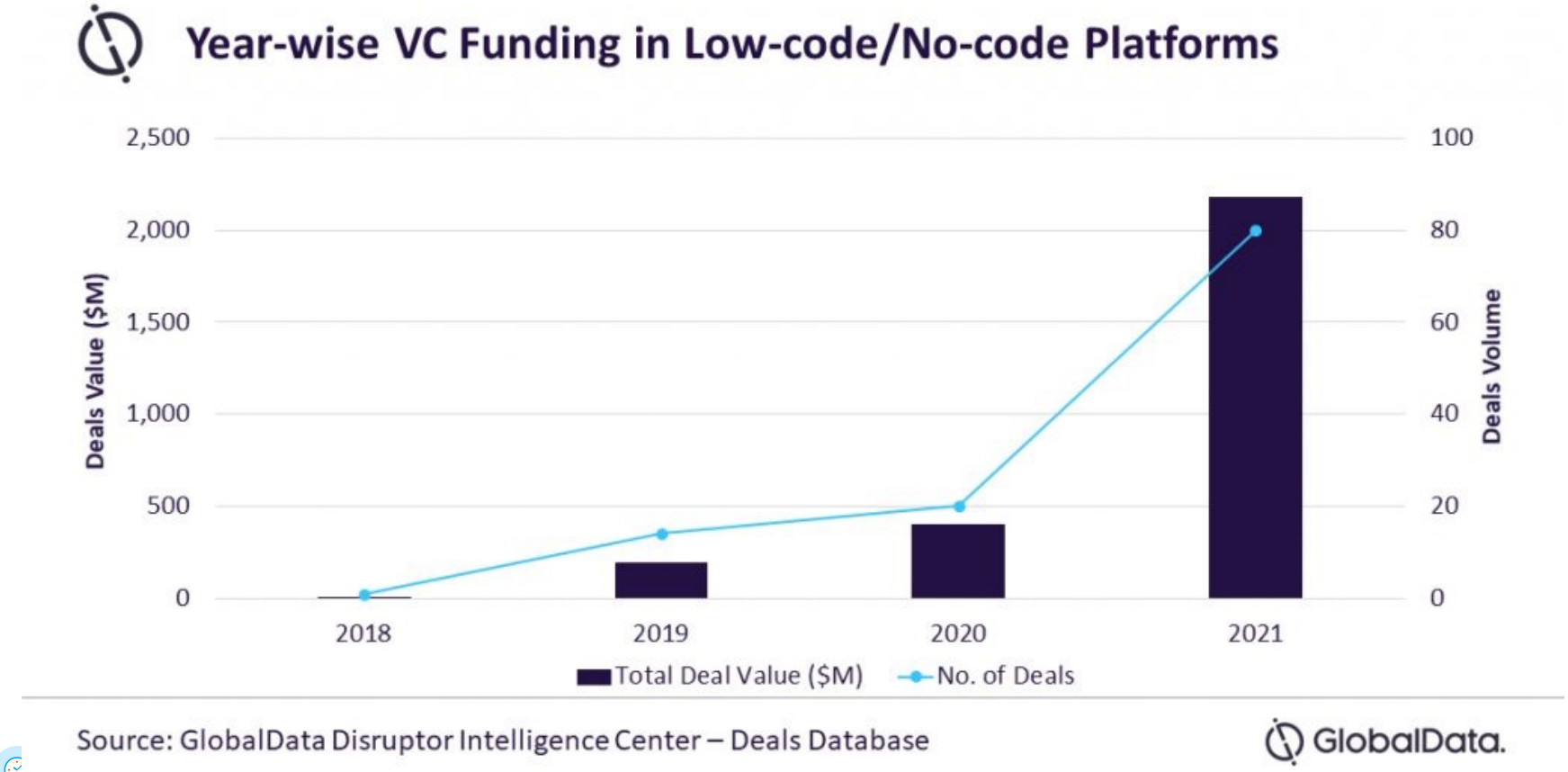
https://www.gartner.com/en/articles/what-s-new-in-the-2023-gartner-hype-cycle-for-emerging-technologies

Lo/no code

What is low code development?

https://www.gartner.com/en/articles/what-s-new-in-the-2023-gartner-hype-cycle-for-emerging-technologies

low-code/no-code Platforms



https://www.enterpriseitworld.com/digital-transformation-drives-fivefold-growth-in-vc-funding-into-low-code-no-code-platforms/

low-code/no-code automation

Customer experience

Pre-built user interface templates to optimize user-friendliness and ensure branding consistency

Visual analytics

Deliver visually appealing descriptive insights for easy consumption using intuitive modeling

Website development

Develop smart web/ mobile apps, automate chatbots, and enable digital analytics management

Marketing and sales automation

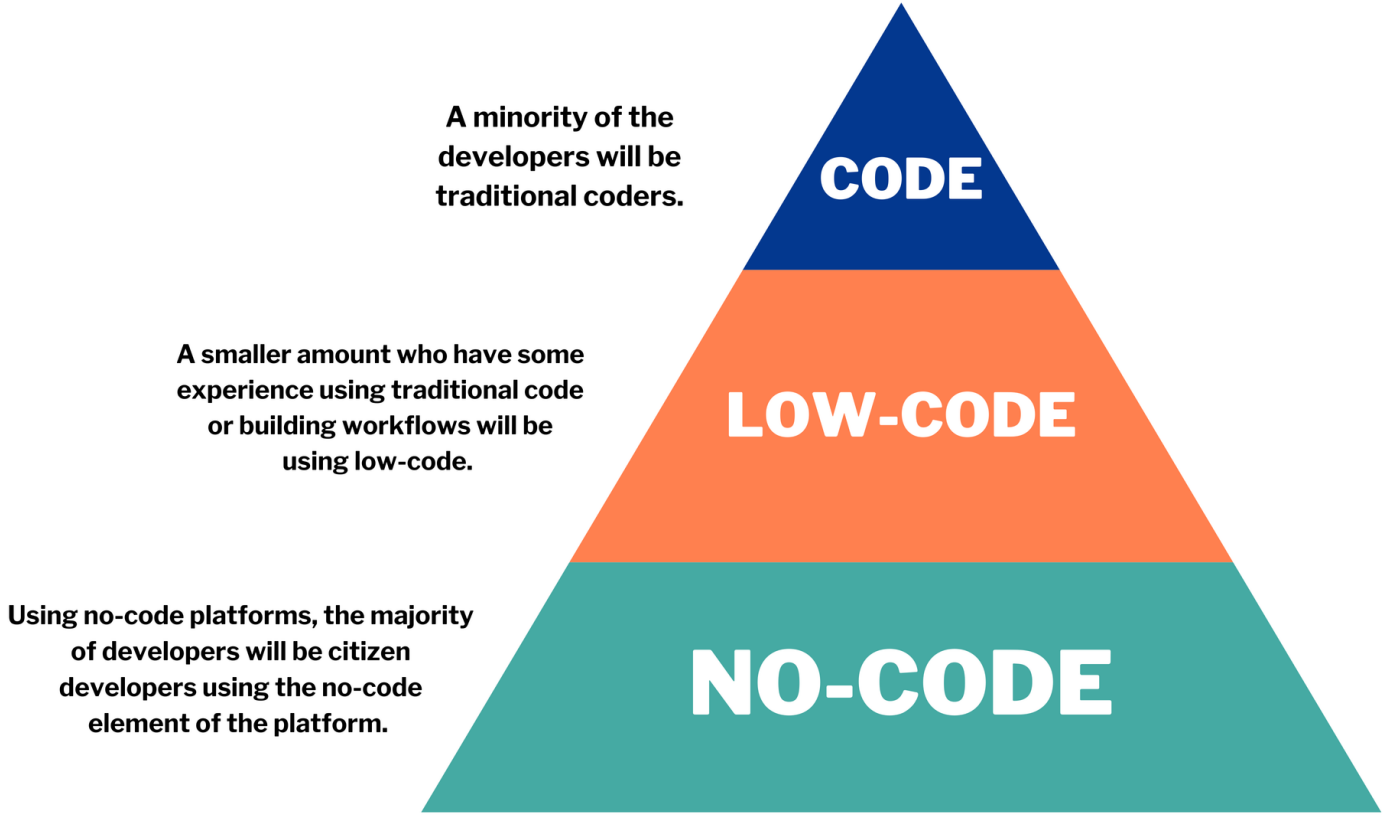
Aid SEO/SEM, and automate personalization, email marketing, and advertising traffic

Managing resources and business transactions

Process appraisals, budget approvals, reservations, order quote creation, and field operations

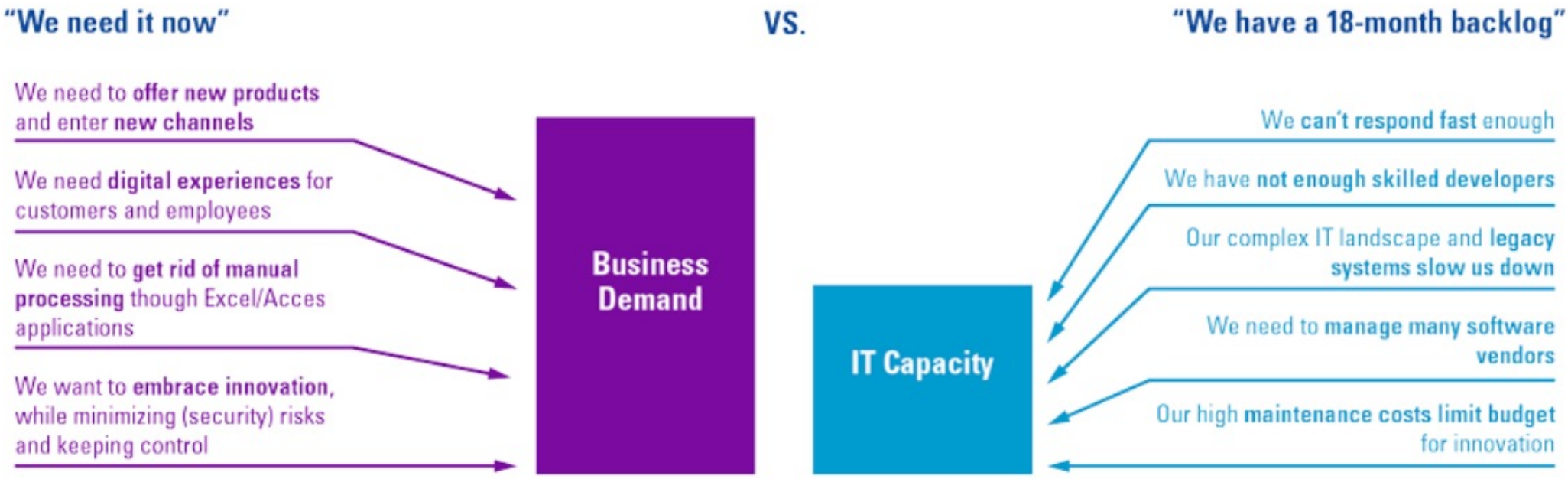
https://www.netscribes.com/low-code-no-code-automation/

How no-code technology is revolutionising digital transformation



https://www.aurora-ecs.com/insights/how-nocode-technology-is-revolutionising-digital-transformation/

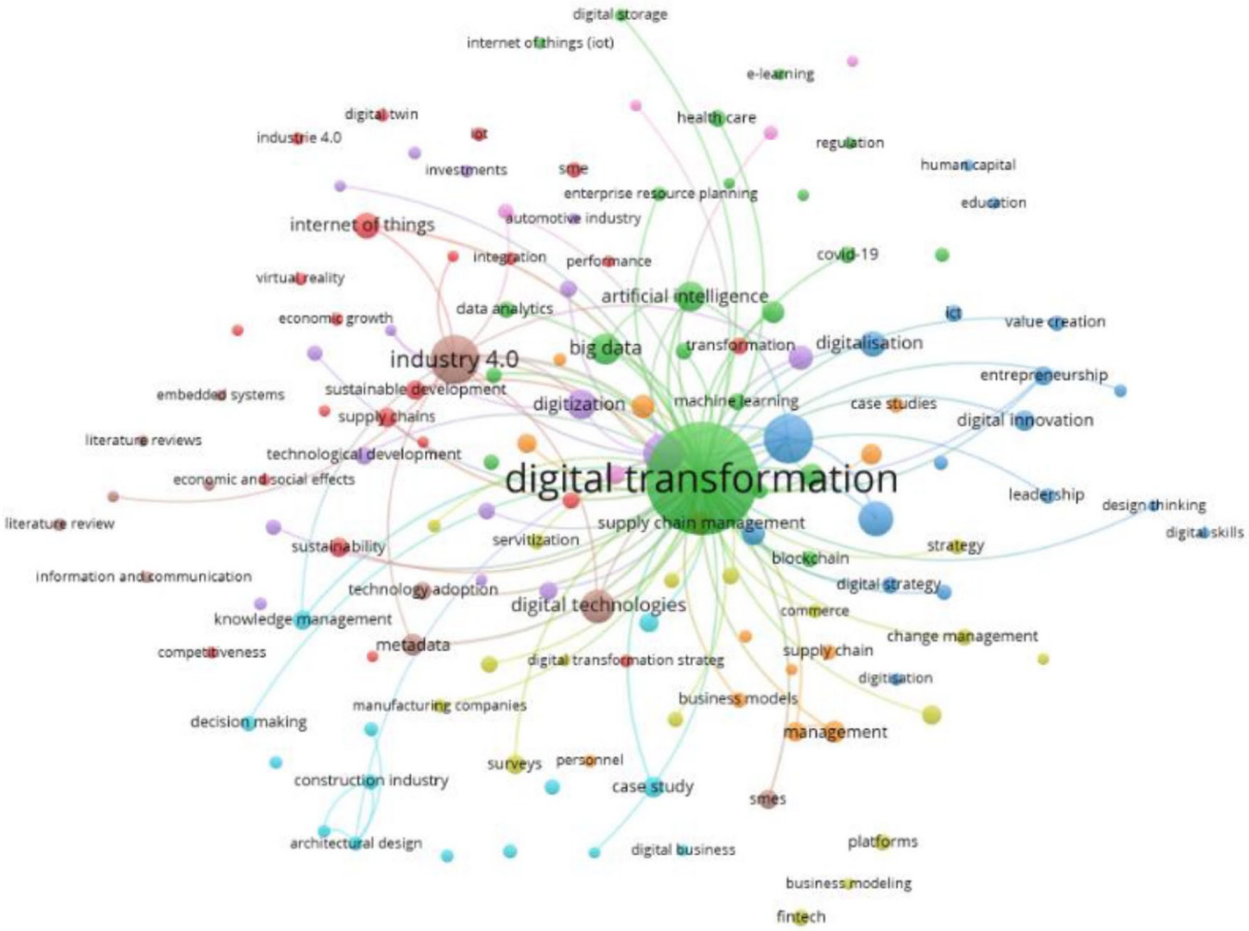
App dev challenges



https://www.compact.nl/en/articles/low-code-empower-the-capability-to-accelerate/

State of the art

Digital Transformation: State of the Art of Research



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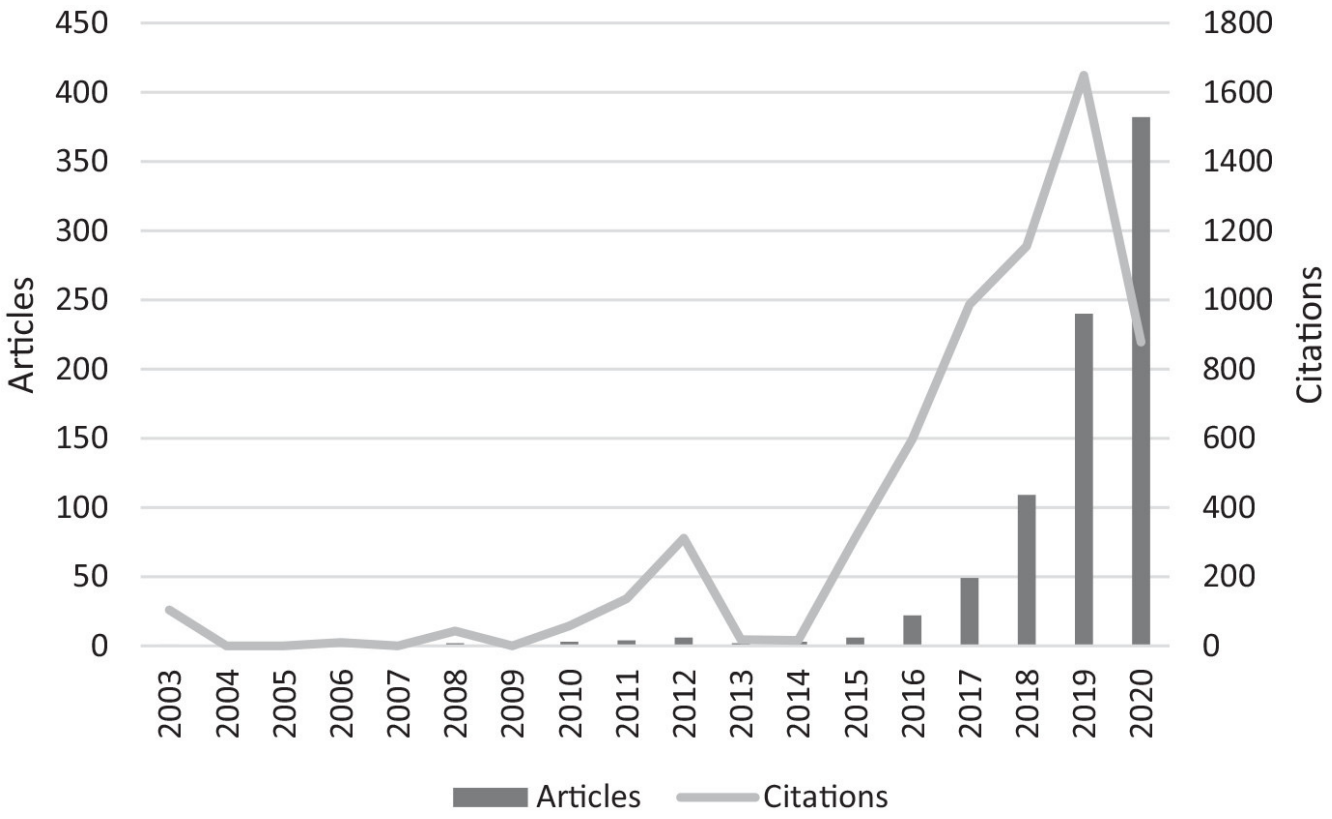

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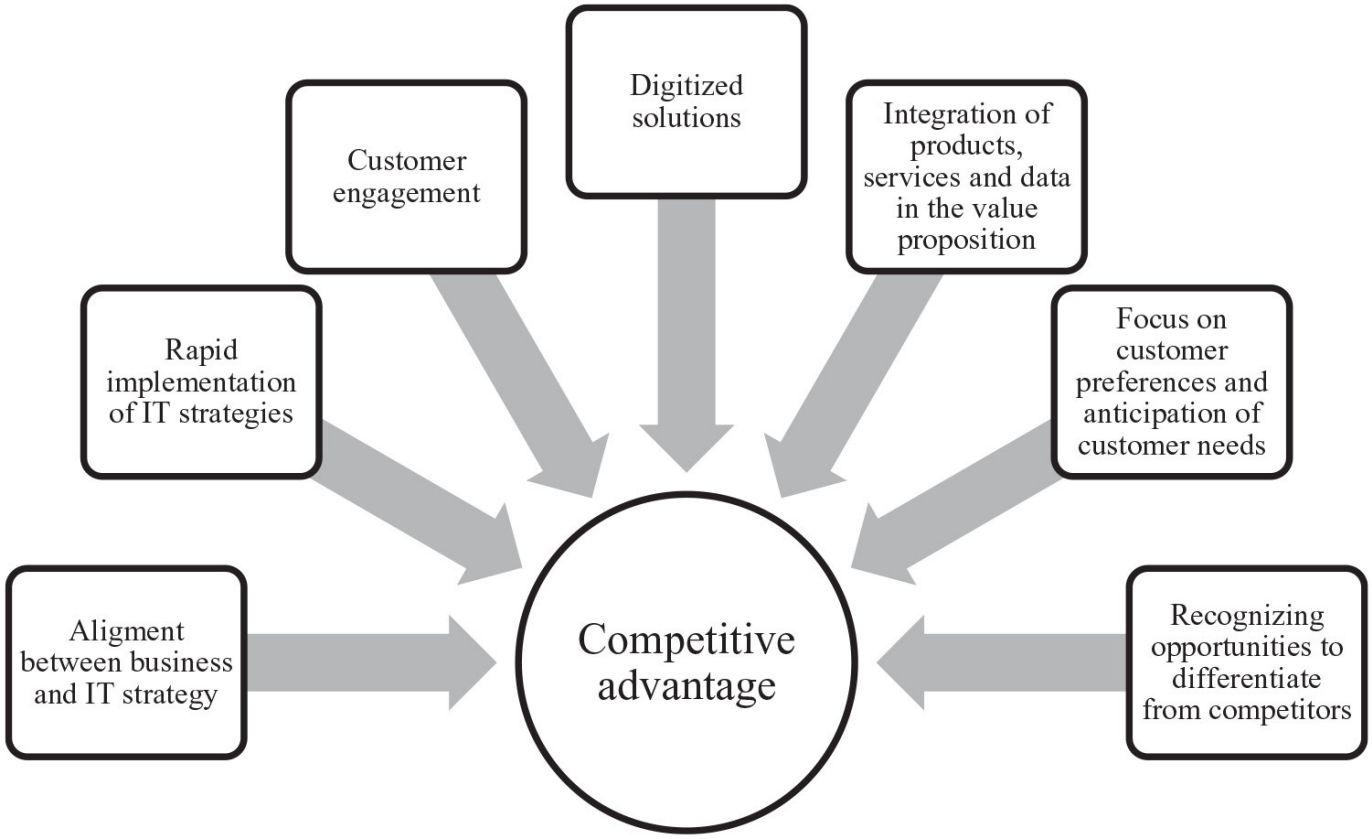

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Main institutional and societal implications regarding digital transformation

- Science becomes more **open, collaborative, and global**
- **Existing rules and norms** within organizations are replaced, complemented, or threatened by novel actors
- **Socio-cultural aspects** influence the way innovations gain legitimacy within organizations
- **Workplaces and employees** may be affected due to increasing interruptions in the virtual working environment
- **Internal security threats emerge**: confidentiality concern, internal **information breaches, disclosure of private information**
- **Risks of automation**: non-automatable tasks

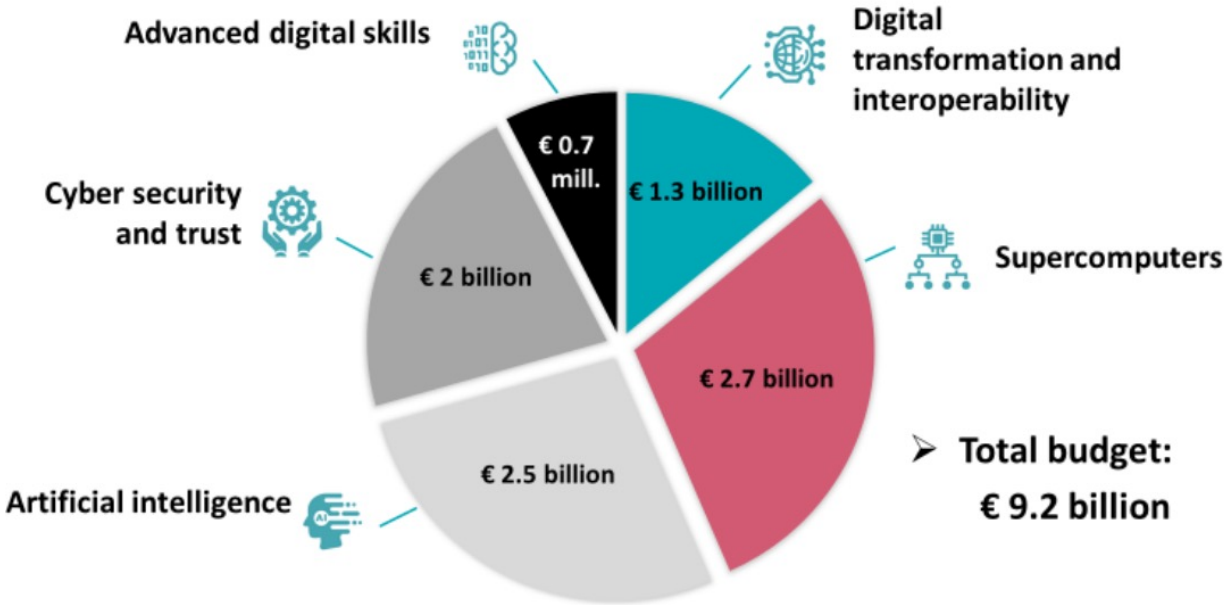
Kraus, S., Jones, P., Kailer, N., Weinmann, A., Chaparro-Banegas, N., & Roig-Tierno, N. (2021). Digital transformation: An overview of the current state of the art of research. Sage Open, 11(3), https://doi.org/10.1177/21582440211047576

A national perspective

Strategy for transformation of Slovakia into a successful digital country

Ministry of Investments, Regional Development and Informatization of the Slovak Republic (2022). 2030 Digital Transformation Strategy for Slovakia, https://mirri.gov.sk/wp-content/uploads/2019/10/SDT-English-Version-FINAL.pdf

EU future challenges, with digital transformation reflected in transport, energy sector, agriculture, health care and culture



- Total budget: € 9.2 billion

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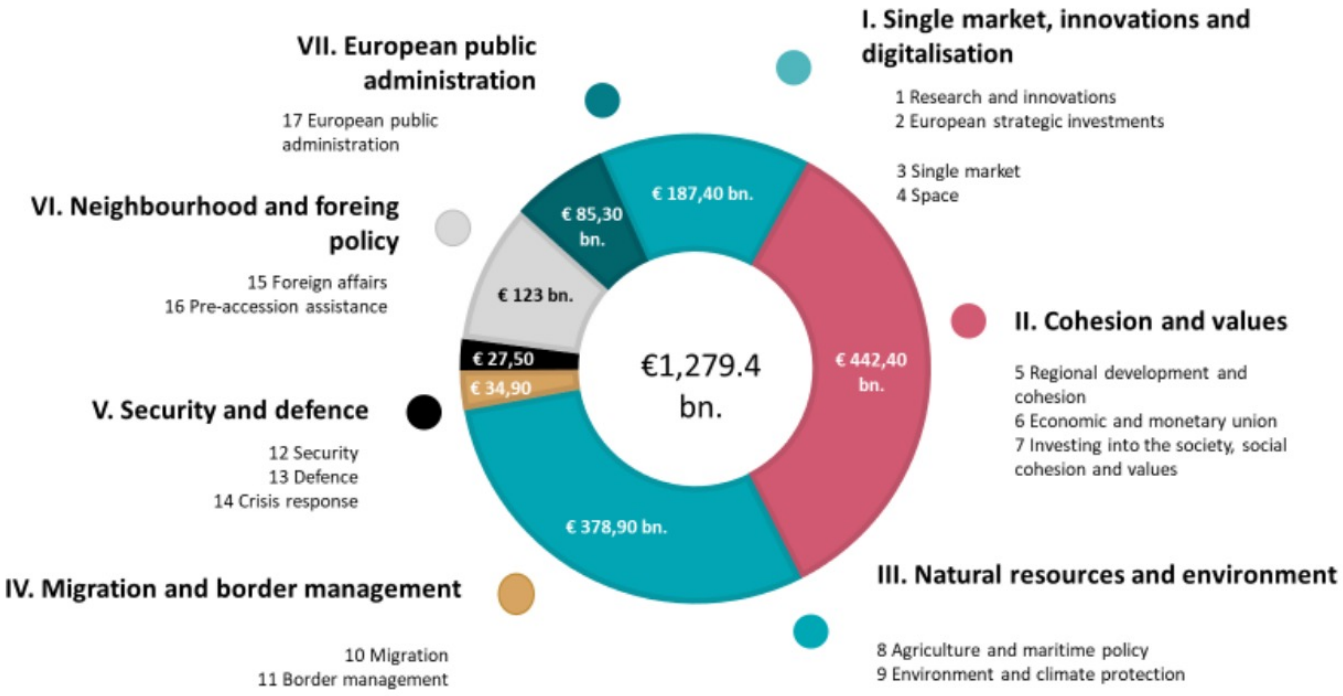

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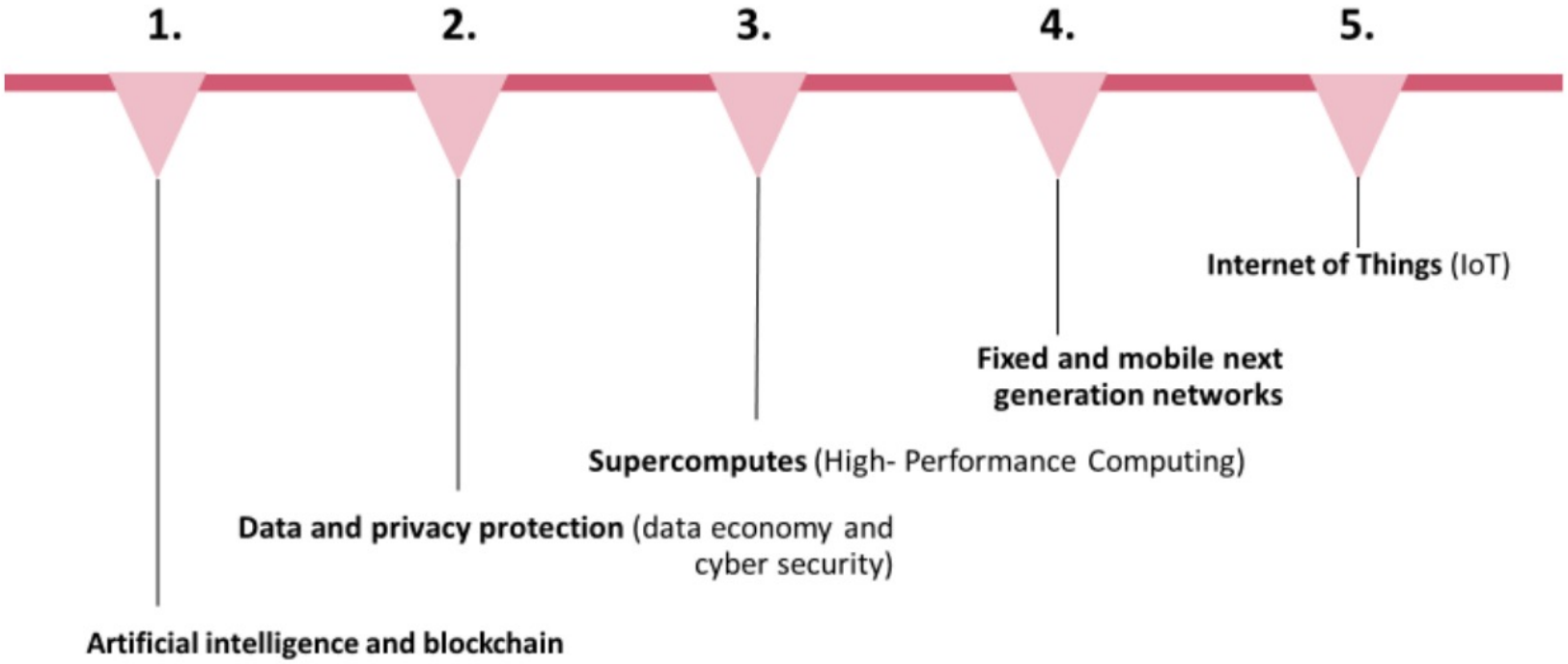
Strategy for transformation of Slovakia into a successful digital country – priority technologies



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Strategy for transformation of Slovakia into a successful digital country – priority technologies - #1 👉AI

- The fundamental risk of AI is the question of ethics, impartiality and transparency
- AI creates systems that are so far capable of fulfilling mainly the strictly specific and rather routine tasks that require a certain level of human intelligence.

Ministry of Investments, Regional Development and Informatization of the Slovak Republic (2022). 2030 Digital Transformation Strategy for Slovakia, https://mirri.gov.sk/wp-content/uploads/2019/10/SDT-English-Version-FINAL.pdf

Strategy for transformation of Slovakia into a successful digital country – priority technologies - #1 👉AI

- The problem of ethics and morality in the short term lies in the fact that these systems are increasingly being deployed to assess individuals' personal data and requests (e.g. in applying for a job, reviewing a loan application, or creating an individual's personal profile based on their habits and behaviours), as well as deciding how the AI will behave in a critical situation, where damage to property or human health cannot be avoided (e.g. in the event of an autonomous vehicle crash).
- This is also related to the questions as to which activities can be left to the AI in terms of ethics, and which should remain explicitly in the hands of the people.

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Attendance